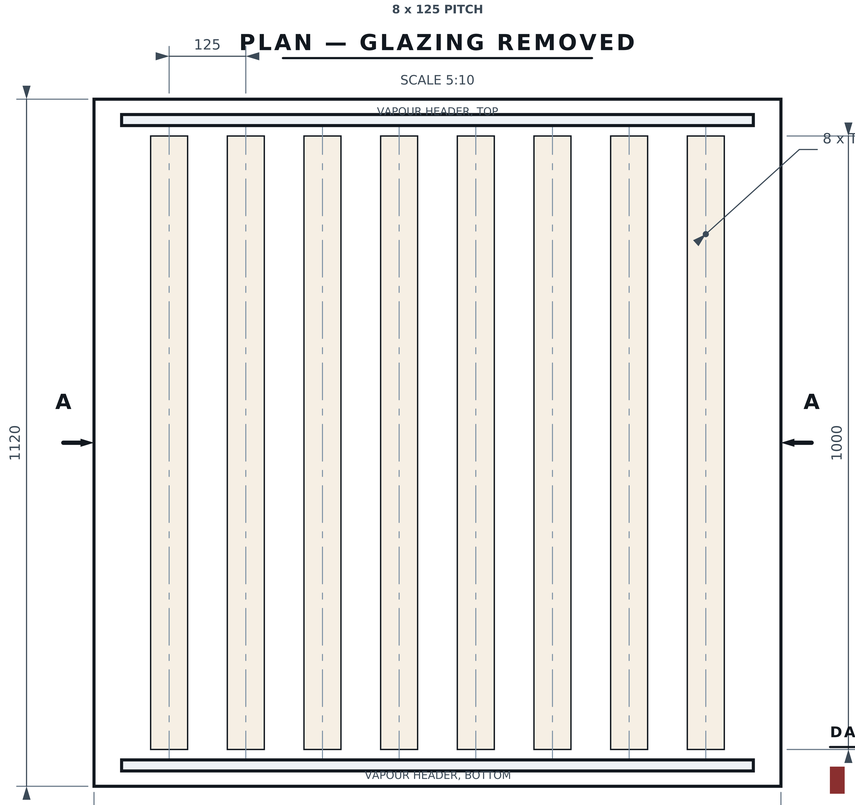
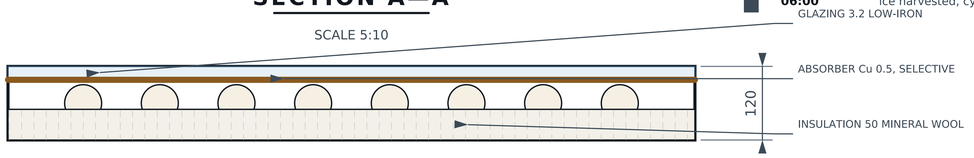
PLAN — GLAZING REMOVED

SCALE 5:10



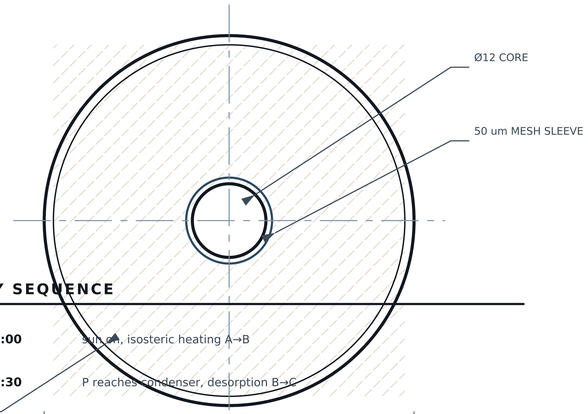
SECTION A—A

SCALE 5:10

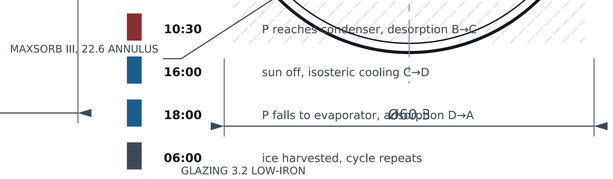


DETAIL B — TUBE SECTION

SCALE 5:1



DAILY SEQUENCE



TUBE DATA	
carbon per tube	0.61 kg
packed volume	16.4 L
fill of available	83 %
annulus	22.6 mm
diffusion time	11.8 min
operating pressure	8.4 bar
stagnation design	**14.0 bar**
hoop at stagnation	26.7 MPa
margin on SS 316L	**4.9 x**

NOTES

1. MATL: tubes and headers SS 316L. Absorber plate Cu, selective coating a 0.95 / e 0.10.
2. STAGNATION IS THE DESIGN CASE. A solar collector will stagnate; at 150 C methanol reaches 14.0 bar.
3. No pressure relief to atmosphere anywhere on this assembly. The tube contains stagnation instead.
4. All joints TIG welded, full penetration. No threaded or elastomer joint on the wetted side.
5. Bake carbon 200 C / 8 h under vacuum before filling; fill under dry nitrogen.
6. Core perforation 2.0 dia, 30 % open, sleeved in 50 um mesh to retain fines.

CRITICAL CHARACTERISTICS			
FEATURE	**NOMINAL**	**TOL**	**CRITICAL TO**
Tube wall	1.5	-0 / +0.15	stagnation containment
Weld, tube to header	full penetration	100 % dye pen	hermeticity
Absorber-to-tube bond	continuous	ultrasonic	collector efficiency
Carbon mass per tube	0.61 kg	±2 %	cycle capacity

INDIA-VC THERMAL HARDWARE — ISO 128 / ISO 2768-mK

TITLE
ADSORBER-COLLECTOR
SS 316L / 8 TUBES OD60.3 / 4.9 kg CARBON / DESIGN 14 bar

SCALE **5:10** — THIRD ANGLE PROJECTION
UNITS **mm**

DRAWING No. **SI-101**
REV **A**
SHEET **2 / 6**

DRAWN	CHECKED	MATERIAL	FINISH	DATE
AI-ASSISTED DRAFT	— PENDING —	SEE BOM	SEE NOTES	2026-09-15